C4.3 State changes on different transactions

The state changes that can be associated with a transaction are determined by:

- the transaction type
- the read response for transactions issued on the AR channel
- whether the master supports an external snoop filter
- whether the master performs speculative reads.

The rules that apply to a master are:

- If a transaction read response has PassDirty asserted, then the cache line must move to a Dirty state. The PassDirty response can be asserted for:
  - ReadNotSharedDirty
  - ReadShared
  - ReadUnique.
- If a transaction read response has IsShared asserted, then the cache line must move to either a Shared state or the Invalid state. The IsShared response can be asserted for:
  - ReadOnce
  - ReadClean
  - ReadNotSharedDirty
  - ReadShared
  - CleanShared.
- A cache line that is in a Unique state is permitted to move to the equivalent Shared state, but this is not expected behavior.
- If an external snoop filter is not supported, a cache line that is in a Clean state can move to the Invalid state.

C4.3.1 State changes associated with a load

No cache line state change is required for the internal action of a load.

C4.3.2 State changes associated with a coherent store

Before carrying out the internal operation of a store to a cache line in shareable memory, the master must ensure that it has permission to store. A master has permission to store if the cache line is in the UniqueClean or UniqueDirty state.

If the master does not have permission to store then it must either:

- Issue a transaction on the AR channel that obtains permission to store, and then perform the store to the cache line. After the store to a cache line, the master must be in the UniqueDirty state. The transactions that obtain permission to store are:
  - ReadUnique
  - CleanUnique
  - MakeUnique.
- Issue a transaction on the AW channel that obtains permission to store and also updates main memory. The transactions that obtain permission to store data and also update main memory are:
  - WriteUnique
  - WriteLineUnique.

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C4.3.3 State changes associated with a main memory update

An update to main memory can be performed when the cache line is in a Dirty state.

When a master is given permission to update main memory, the earliest the associated write transaction can occur is the cycle after the **RVALID**/**RREADY** handshake in which **RLAST** is asserted for the transaction that gave permission to update main memory.

An update to main memory is performed using a WriteBack or WriteClean transaction.

After an update to main memory, the cache line must be in a Clean or Invalid state.

If an external snoop filter is supported, then the following restrictions apply:

- after a WriteBack transaction, the cache line must be in the Invalid state
- after a WriteClean transaction, the cache line must be in a Clean state.

C4.3.4 State changes associated with cache maintenance operations

The cache maintenance transactions are:

- CleanShared
- CleanInvalid
- MakeInvalid.

Before issuing a cache maintenance transaction, the master must ensure that:

- for CleanShared, the cache line must be in a Clean or Invalid state
- for CleanInvalid and MakeInvalid, the cache line must be in the Invalid state.

Note

A cache maintenance transaction does not change the cache line state.

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